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Sotira — 6.53 MWp east–west, with a 22.9 MWh battery sized to the site

6.53 MWP | YIELD MODELLED ON PVGIS 5.3 | EAST–WEST 12.5° | 22.9 MWH · THREE-AND-A-HALF HOUR | SOTIRA, FAMAGUSTA DISTRICT | ABOVE THE 5 MWP CEILING

A 6.53 MWp east–west park at Sotira in the Famagusta District, on the geometry confirmed with the site owner: 12.5° tilt, 1,250 mm low-edge clearance, 10.603 m shed pitch, 359 strings of 28 bifacial Jinko 650 W modules into twenty Huawei 330 kW inverters. It is the plant whose TMY and bifacial rear-side model set the south-facing yield used on every other page in this pack. Ready-to-build is costed at €350,000 per megawatt. It is also larger than the envelope in M2's brief.

6.53 MWp	6.53 / 22.9	€10.76m	€1,199,100	11.1%
East–west 12.5°, bifacial	MW / MWh LFP, three-and-a-half hour	All-in, incl. RTB at €350k/MW	Year 1 EBITDA, unlevered	Year 1 yield on cost

The yield is measured, not assumed — and the south-facing number is taken from here

Specific yield on this plant is **1,806 kWh/kWp · yr**. That is not a worse site; it is a different geometry. An east–west array at 12.5° gives up about 9% against a south-facing array at the same latitude, and the 88% ground coverage ratio that makes the layout fit the Poleodomia envelope also closes off the ground behind the modules, so bifacial gain is 3.6% rather than the 9.3% a 45% GCR south-facing array achieves at the same albedo. The south-facing figure of **2,194 kWh/kWp** used for Agios Theodoros, the Margí expansion and the 1 MWp block is this site's PVGIS south-optimum 31° front (2,006) with that same rear-side model. We have set both numbers out rather than smoothed them.

Configuration at this latitude	kWh/kWp · yr	Note
PVGIS south-facing optimum, 31°, front only	2,006	Theoretical front-side ceiling for the location
Same, plus bifacial gain at GCR 0.45 / albedo 0.25	**2,194**	**+9.3% — the south-facing figure used in this pack**
Same modules, south 12°, front only	1,917	Orientation comparison
This plant, east–west 12.5°, front only	1,743	−9.1% for the east–west geometry
This plant with bifacial gain at albedo 0.25	1,806	+3.6%, limited by 88% ground coverage

Source: JRC PVGIS 5.3 (SARAH3, 2005–2023) at 35.01084°N, 33.94224°E, with a two-dimensional rear-side model (pvlib 0.15 infinite sheds, Hay-Davies) on the PVGIS TMY. Bifaciality 0.85. Front-side loss stack −11.2%, performance ratio 88.8%. This is an indicative energy note and not a bankable PVsyst P50/P90; a PVsyst run against the PAN and OND files remains the document a lender will want, and cable ohmic losses are not yet in the stack — each further 1% removes about 118 MWh a year.

Why the battery here is three and a half hours, not four

The four-hour ratio in Document 1 is not a house rule, it is the point at which the curtailed energy runs out. Lower specific yield means less curtailed energy per megawatt-peak, so that point arrives sooner. At Sotira the battery saturates at 22.9 MWh; a 26.1 MWh four-hour system would deliver **exactly the same energy** for a further €414,900 of capital. Sizing to the site rather than to the ratio is worth 0.3 points of yield on the all-in cost below.

Battery	Capacity	Capex	Delivered to evening market	Capture	Battery revenue
Two hours	13.07 MWh	€1,659,600	4,188 MWh	54.6%	€732,900
Three hours	19.60 MWh	€2,489,500	6,282 MWh	81.9%	€1,099,300
Three and a half hours — recommended	**22.87 MWh**	**€2,904,400**	**6,532 MWh**	**85.2%**	**€1,143,200**
Four hours — pack default	26.14 MWh	€3,319,300	6,532 MWh	85.2%	€1,143,200

At 65% curtailment. The fourth hour returns nothing at this site.

Year one

Energy and revenue	MWh	€/MWh	€
Gross generation at 1,806 kWh/kWp	11,800	—	—
Exported directly (35%)	4,130	127	524,500
Recovered by battery, sold at night	6,532	175	1,143,200
Curtailed, not recovered	1,138	—	—
Gross energy revenue	**10,662**	**156**	**1,667,700**
Aggregator and offtake fee, 10%			(166,800)
Operating cost — O&M, insurance, land			(301,900)
Year 1 EBITDA, unlevered			**1,199,100**

All-in cost to a buyer, ex VAT	€
Ready-to-build — 6.534 MWp at €350k/MW	2,286,900
PV EPC — 6.534 MWp at €0.72/Wp	4,704,500
BESS EPC — 22.87 MWh at €127k/MWh	2,904,400
Development, grid, permits, contingency	866,300
Total investment	**10,762,100**
All-in cost per MWp	1,647,100
Year 1 yield on cost	11.1%
Simple payback, unlevered	9.0 years
Realised average price, all energy sold	€156/MWh

Same EPC and battery rates as every other asset in this pack. RTB is the Cyprus ready-to-build rate of €350,000/MW. Land lease terms sit in the data room and are not double-counted here.

Sensitivity to curtailment

Curtailment	Exported	Recovered by battery	Gross revenue	Year 1 EBITDA	Yield on cost	Payback
35%	7,670 MWh	3,517 MWh	€1,589,700	€1,128,800	10.5%	9.5 yr
50% — 2025 national actual 47.4%	5,900 MWh	5,025 MWh	€1,628,700	€1,164,000	10.8%	9.2 yr
65% — 2026 national actual, base case	**4,130 MWh**	**6,532 MWh**	**€1,667,700**	**€1,199,100**	**11.1%**	**9.0 yr**
75%	2,950 MWh	7,329 MWh	€1,657,200	€1,189,600	11.1%	9.0 yr

Not modelled, and to the upside: an east–west array produces a flatter daily curve than a south-facing one, shifting output out of the midday hours where curtailment is concentrated and into the morning and late-afternoon shoulders where the grid still accepts it. That should raise the exported share above the 35% national average assumed here. Quantifying it needs half-hourly modelling against the site's own dispatch signal, which we have not yet done and have therefore taken no credit for.

Two things M2 should know before spending time on this one. At 6.53 MWp Sotira is above the 5 MWp ceiling in the original brief and cannot be trimmed without redoing the layout, which is fitted to the Poleodomia envelope at 6,534 kW. Commercial terms beyond the €350k/MW RTB rate used here are not a signed SPA — this document describes an engineering case with that rate loaded, not a closed transaction. It is in the pack because the yield work behind it is the most rigorous we have, and because if M2's envelope is indicative rather than firm it is the largest single asset we can offer.

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Basis. Wholesale prices from published TSOC day-ahead market files, 1 Oct 2025 – 11 Feb 2026 (134 trading days, 6,432 half-hourly prints): daytime average €127/MWh, evening peak (17:00–21:00) average €183/MWh. This pack takes €175/MWh for evening capture, €8/MWh below the measured average. Curtailment base case of 65% is the published national rate for January–May 2026 (162 GWh curtailed); 2025 full year was 47.4% (306 GWh, CyprusGrid). South-facing new-build yield is 2,194 kWh/kWp · yr (PVGIS 5.3 south 31° front 2,006 at the Sotira TMY, plus +9.3% bifacial at albedo 0.25, GCR 0.45, φ 0.85, 680 Wp n-type). Margí operating uses metered 2,100. Sotira east–west is 1,806. PVsyst before commitment. Battery round-trip efficiency 87.8% AC–AC; effective capture of curtailed energy 85%, being daily-shape collection multiplied by round-trip efficiency. EPC rates €0.72/Wp (PV, turnkey) and €127,000/MWh (BESS, ex VAT) per the Lighthief EPC workbook v4, February 2026, and are indicative pending final specification and equipment pricing at the date of order. Ready-to-build site cost is €350,000/MW where an acquisition is required. Revenues are stated after a 10% aggregator and offtake fee on merchant energy and before Cyprus corporate income tax, which rose from 12.5% to 15% on 1 January 2026. Figures are unlevered and exclude VAT. Merchant exposure to a day-ahead market with under two years of published price history is the single largest uncertainty in this case; no merchant revenue is contracted.